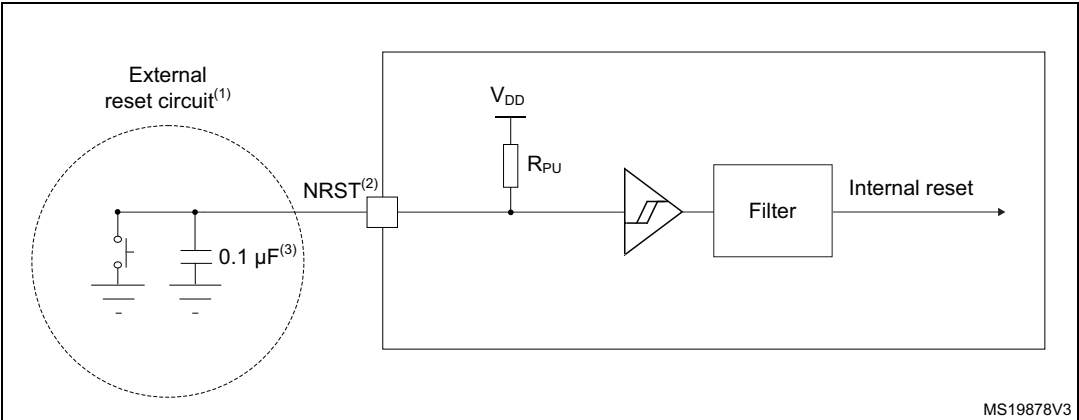


Figure 19. Recommended NRST pin protection



1. The reset network protects the device against parasitic resets.
2. The user must ensure that, upon power-on, the level on the NRST pin can exceed the minimum $V_{IH(NRST)}$ level specified in [Table 52: NRST pin characteristics](#). Otherwise, the device does not exit the power-on reset. This applies to any NRST configuration set through the NRST_MODE[1:0] bitfield, the GPIO mode inclusive.
3. The external capacitor on NRST must be placed as close as possible to the device.

5.3.15 Extended interrupt and event controller input (EXTI) characteristics

Table 53. EXTI input characteristics

Symbol	Parameter ⁽¹⁾	Conditions	Min	Typ	Max	Unit
PLEC	Pulse length to event controller	-	20	-	-	ns

1. Specified by design. Not tested in production.

5.3.16 Analog-to-digital converter characteristics

Unless otherwise specified, the parameters given in [Table 54](#) are preliminary values derived from tests performed under ambient temperature, f_{CLK} frequency and V_{DDA} supply voltage conditions summarized in [Table 23: General operating conditions](#).

Note: It is recommended to perform a calibration after each power-up.

Table 54. ADC characteristics

Symbol	Parameter ⁽¹⁾	Conditions	Min	Typ	Max	Unit
V_{DDA}	Analog supply voltage	-	2.0	-	3.6	V
V_{REF+}	Positive reference voltage	-	2	-	V_{DD}	V
f_{ADC}	ADC clock frequency	-	0.14	-	35	MHz
f_s	Sampling rate	12 bits	-	-	2.50	MSps
		10 bits	-	-	2.92	
		8 bits	-	-	3.50	
		6 bits	-	-	4.38	